Gantry crane Specification

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Revision	A
Status	APPROVED
Classification	CONFIDENTIAL
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Approved by	Petra Valcourt
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1. Purpose

This specification defines the performance, interface and verification requirements for the gantry crane (SY-GCR) of the Shipyard Slipways on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-SY-03 (WBS 1.7.3), owned by Quell Palin.

The prime supplier for flight hardware is Kuat Drive Yards. Responsible engineer: Barrin Penrose.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-SY-DWG-0202	Gantry crane general arrangement drawing
SHV-SY-BOM-0302	Gantry crane bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-SY	Interface Control Document: Power Distribution / Shipyard Slipways
SHV-ICD-HB-SY	Interface Control Document: Docking & Hangar Bays / Shipyard Slipways
SHV-ICD-SF-SY	Interface Control Document: Superstructure & Frame / Shipyard Slipways

4. Description

The shipyard builds and refits freighters, liners and survey ships on pressurized slipways and dry docks, served by gantry cranes, a coating shop and outfitting quays.

The gantry crane is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
SY-GCR-R01	The gantry crane shall provide a lift capacity of 3,800 t.	3,800 t	±1%	Analysis
SY-GCR-R02	The gantry crane shall provide a crane span of 168 m.	168 m	±2.5%	Inspection

5.1 General requirements

ID	Requirement	Verification
SY-GCR-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
SY-GCR-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
SY-GCR-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
SY-GCR-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
SY-GCR-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the SY test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-06-25	Initial release.	Barrin Penrose